



The Missing Block

Quick facilitation notes & answer key

Goal

Students fix a MISSING-block bug by writing the block a script needs (C3.1 write + C3.2 alter). The goal is 'meet on 3'; Bit is correct, but Pixel has only one 'back 1' (lands on 4) and the dashed slot shows a block is missing. Students work out the remaining distance and add 'back 1'. Adding a block to complete a script is a key debugging-and-writing skill.

Ontario curriculum (Grade 2 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: a star on square 3; run Pixel's one move, then count the remaining squares to the star to find the missing block.

Run it (about 20 min)

1. I DO: state the goal — 'meet on 3.' Run Bit: 0 forward 3 → 3 (correct).
2. WE DO: run Pixel with its one move: 5 back 1 → 4. 'Pixel stops on 4 — one square short. A block is missing.'
3. YOU DO: students count 4 → 3 (one more back), write 'back 1' in the dashed box, and confirm Pixel reaches 3.
4. Connect: 'If a script stops short, add the block that finishes the path.'

Answer key (modelled by the run-gate)

Ex 1 — Pixel with one move: 5 → back 1 → 4. It does NOT reach the goal 3 (one short).

Ex 2 — Pixel is on 4 and needs 3, so it needs 1 more square back; the missing block is 'back 1'.

Ex 3 — add 'back 1' in the dashed box → Pixel: 5 → back 1 → 4 → back 1 → 3. Now both meet on 3.

Fix note: a single 'back 1' completes it; a child who writes 'back 1' (or an equivalent that lands on 3) is correct.

Watch for & differentiate

Common error: adding a forward block — Pixel is past the goal's side, so it needs to keep going back.

Common error: over-adding (back 2) so Pixel overshoots to 2 — count the remaining distance carefully (just 1).

Simplify: put a finger on Pixel's current spot (4) and hop to the star (3) — one hop back.

Extend: ask what would be missing if Pixel's first move were 'back 2' instead (nothing — it would already be on 3).

Success indicator: the child adds one 'back 1' and confirms both sprites meet on 3.